

EWMA Volatility Model

Portfolio VaR Engine — Model Documentation

Risk Modelling — Trade Republic

April 2026

1 Overview

The Exponentially Weighted Moving Average (EWMA) engine estimates a time-varying portfolio volatility using an exponential weighting scheme and derives parametric VaR and CVaR from either the Normal or Student- t distribution. This is the RiskMetrics-style approach popularised by J.P. Morgan (1994).

Engine key: 'ewma'

Sources:

- `risk_analytics/risk_engines.py`, lines 190–225 (VaR/CVaR formulas)
- `risk_models/support_models.py`, lines 12–30 (`ewma_volatility`)

2 EWMA Volatility

2.1 Definition

Given a portfolio return series $\{r_t^{pf}\}$ and a decay parameter $\lambda \in (0, 1)$, the EWMA variance estimate at time t over a window of T observations is:

$$\hat{\sigma}_t^2 = \sum_{k=0}^{T-1} \lambda^k (1 - \lambda) (r_{t-k}^{pf})^2 \quad (1)$$

The EWMA volatility is $\hat{\sigma}_t = \sqrt{\hat{\sigma}_t^2}$.

Weighting scheme. The weight assigned to observation k periods ago is $w_k = \lambda^k (1 - \lambda)$, with $\sum_{k=0}^{T-1} w_k \approx 1$ for large T . More recent observations receive exponentially higher weight, allowing the model to respond quickly to volatility regimes. The default $\lambda = 0.94$ corresponds to the RiskMetrics specification for daily data.

2.2 Implementation

The implementation in `support_models.ewma_volatility()` pre-computes the weight vector once and applies it via a sliding window:

$$\hat{\sigma}_t^2 = (\mathbf{w}^\top \mathbf{r}^2)_t, \quad \mathbf{w} = [\lambda^{T-1}(1 - \lambda), \dots, \lambda^0(1 - \lambda)] \quad (2)$$

where $\mathbf{r}^2$ denotes element-wise squared returns in the window.

3 VaR and CVaR Formulas

3.1 Normal Distribution

The distribution of r_t^{pf} is assumed $\mathcal{N}(\hat{\mu}, \hat{\sigma}_t^2)$, where $\hat{\mu}$ is the MLE sample mean.

Value-at-Risk:

$$\widehat{\text{VaR}}_{\alpha}^N = \hat{\mu} - \hat{\sigma}_t \cdot \Phi^{-1}(\alpha) \quad (3)$$

Since $\alpha < 0.5$, we have $\Phi^{-1}(\alpha) < 0$, so $\widehat{\text{VaR}}_{\alpha}^N > 0$ (a positive loss).

Conditional VaR (Expected Shortfall):

$$\widehat{\text{CVaR}}_{\alpha}^N = -\hat{\mu} + \hat{\sigma}_t \cdot \frac{\phi(\Phi^{-1}(\alpha))}{\alpha} \quad (4)$$

where $\phi(\cdot)$ is the standard Normal PDF and $\Phi^{-1}(\cdot)$ is the Normal quantile function. This is the closed-form CVaR for a Gaussian distribution.

3.2 Student- t Distribution

The distribution of r_t^{pf} is assumed Student- t with ν degrees of freedom, location $\hat{\mu}_t$, and scale $\hat{s}_t$, all estimated via MLE.

Value-at-Risk:

$$\widehat{\text{VaR}}_{\alpha}^t = \hat{\mu}_t - \hat{\sigma}_t \sqrt{\frac{\nu-2}{\nu}} \cdot t^{-1}(\alpha; \nu) \quad (5)$$

The $\sqrt{(\nu-2)/\nu}$ term converts the MLE scale parameter to the standard deviation of the fitted t -distribution.

Conditional VaR (Expected Shortfall):

$$\widehat{\text{CVaR}}_{\alpha}^t = -\hat{\mu}_t - \frac{\hat{\sigma}_t}{\alpha(1-\nu)} \left(\nu - 2 + [t^{-1}(\alpha; \nu)]^2 \right) \cdot f_t(t^{-1}(\alpha; \nu); \nu) \quad (6)$$

where $t^{-1}(\alpha; \nu)$ is the α -quantile of the standard Student- t distribution with ν degrees of freedom, $f_t(\cdot; \nu)$ is its PDF, and $\nu \geq 3$ is enforced (a floor of 3 is applied if the MLE estimate falls below 2).

3.3 Holding-Period Scaling

$$\widehat{\text{VaR}}_{\alpha}^{(H)} = \widehat{\text{VaR}}_{\alpha}^{(1)} \cdot \sqrt{H}, \quad \widehat{\text{CVaR}}_{\alpha}^{(H)} = \widehat{\text{CVaR}}_{\alpha}^{(1)} \cdot \sqrt{H} \quad (7)$$

4 Parameters

Parameter	Type	Default	Description
window	int	250	Lookback window for volatility estimation
decay	float	0.94	EWMA decay λ (RiskMetrics default)
distr	str	'norm'	Distribution: 'norm' or 't'
qtls	list	[0.05, 0.01, 0.001]	Tail probability levels
holding_period	int	1	Forecast horizon in days
verbose	bool	True	Print model setup information

5 Advantages and Limitations

Advantages	Limitations
Adapts quickly to volatility changes	No asymmetric (leverage) effects
Closed-form VaR and CVaR	Assumes a single time-varying scale
Computationally inexpensive	Parametric tail; may underestimate extreme losses
RiskMetrics industry standard	EWMA variance ignores the mean

6 Usage Example

```
from risk_analytics import risk_engines as re

var_ewma = re.portfolio_var(
    rets_orig = rets,
    wgts       = weights,
    engine     = 'ewma',
    fmt_engine = {
        'window'       : 250,
        'decay'        : 0.94,
        'distr'         : 'norm',
        'qtls'          : [0.05, 0.01, 0.001],
        'holding_period' : 1,
    }
)
```